

# MEDI-SAARTHI

## PRODUCT REQUIREMENTS DOCUMENT (PRD)

**Version:** 1.0 | **Platform:** Web Application / PWA | **SIH PS:** SIH26133

**Tagline:** From fragmented care to connected healthcare

### 1. PRODUCT OVERVIEW

**Problem:** Rural and underserved patients may face long travel distances, limited specialists, unclear facility/service availability, fragmented information, delayed referrals, manual data entry, poor follow-up and connectivity/language barriers.

**Solution:** Medi-Saarthi is a healthcare coordination platform connecting the patient's journey from sub-centre/PHC through referral, treatment and follow-up. It acts as a coordination layer rather than replacing existing public-health systems.

### 2. VISION, GOALS & BOUNDARIES

**Vision:** Ensure that no rural patient gets lost between healthcare facilities because of a broken referral, unavailable service, missing information or missed follow-up.

**Goals:** reduce unnecessary travel; improve referral completion; show facility readiness; reduce frontline-worker data-entry burden; maintain continuity; support low connectivity; track missed referrals; provide administrator visibility.

**Non-goals:** autonomous diagnosis; replacing doctors; creating beds/specialists/medicines/ambulances; replacing ABDM or eSanjeevani; operating an independent ambulance network.

**Medi-Saarthi coordinates within available healthcare capacity.**

### 3. TARGET USERS

**Frontline workers:** register patients, capture information, use triage support, check readiness, create referrals and manage follow-ups.

**Facility staff:** receive referrals, accept/reject, update availability, confirm arrival and care completion.

**Doctors:** review relevant information, provide care, complete referrals and recommend follow-up.

**District administrators:** monitor referrals, bottlenecks, readiness, missed referrals and quality indicators.

### 4. CORE PATIENT JOURNEY

**Patient → Assisted Intake → Rule-Based Triage → Pulse-Check → Facility Selection → Create Referral → Accept/Reject → Arrive → Care Completed → Follow-up → Closed**

### 5. MVP

The MVP proves the end-to-end referral journey: **Worker Login → Patient Registration → Voice/Text assisted capture → Rule-based triage → Facility availability → Referral creation → Facility acceptance → Patient arrival → Completion → Follow-up task.**

**Demo failure case:** Facility rejects a referral because the required service is unavailable. The rejection reason is recorded and the next action/follow-up is created.

### 6. KEY FEATURES

**Assisted Intake:** Voice-to-text and OCR reduce repeated manual entry; extracted information must be reviewed by the worker.

**Rule-Based Triage:** Suggests routine care, teleconsultation, referral or urgent escalation. It does not autonomously diagnose.

**Pulse-Check:** Shows doctor/service/slot/resource readiness with verification timestamp and stale-data warning.

**Referral Guard:** Tracks **Create → Accept → Arrive → Complete → Close**; rejected/missed referrals remain auditable.

**Identity Resilience:** Supports an offline-first episode ID approach where normal identity/connectivity is unavailable.

**Follow-up:** Generates tasks for missed referrals and scheduled follow-up.

**Emergency support:** Prepares structured information for approved emergency/dispatch integrations; it does not create an ambulance network.

## 7. FRONTLINE WORKER DASHBOARD

Quick actions: **Register Patient, New Referral, Check Facility Availability, View Referrals, Follow-up Tasks.** Show counts for pending, accepted, completed and missed referrals.

## 8. PATIENT REGISTRATION

MVP fields: name, age, gender, village, contact/guardian contact, basic symptoms, urgency indicators and optional existing health ID. Avoid unnecessary mandatory fields.

## 9. REFERRAL GUARD

**State machine:** **CREATED** → **ACCEPTED** → **ARRIVED** → **COMPLETED** → **CLOSED**

**Alternative:** CREATED → REJECTED → NEXT ACTION. Record rejection reason and referral events for auditability.

## 10. PULSE-CHECK

Facility staff can maintain service status such as Available, Unavailable or Stale. Store last-verified time and, where useful, who confirmed it. Never present stale data as guaranteed availability.

## 11. FOLLOW-UP & ADMIN

Missed/incomplete referrals can generate follow-up tasks. Admin dashboard can show total referrals, completed, pending, missed, completion rate, referrals by facility, rejection reasons, average referral time, missed follow-ups and readiness indicators.

## 12. OFFLINE-FIRST

Essential capture continues when connectivity is unavailable using browser storage/IndexedDB and a synchronization queue. When connectivity returns, queued events sync with the backend. Sensitive local data should be minimized and protected.

**Offline demo:** **Register patient offline** → **save locally** → **reconnect** → **synchronize.**

## 13. TECH STACK

**Frontend:** React + Vite + TypeScript + Tailwind CSS + PWA

**Backend:** Node.js + Express.js + TypeScript

**Database:** PostgreSQL

**Optional:** Redis for frequently changing availability/cache/queue information

**Voice:** Web Speech API or suitable speech-to-text service | **OCR:** Tesseract.js prototype | **Maps:** Leaflet + OpenStreetMap

**Interoperability:** FHIR/ABDM-compatible API design | **Notifications:** in-app first, SMS/IVR-ready architecture | **Deployment:** Vercel + Render + managed PostgreSQL (prototype option)

## 14. HIGH-LEVEL ARCHITECTURE

Users → React/PWA → REST/HTTPS API → Node.js/Express → PostgreSQL + optional Redis → External adapters (FHIR/ABDM, SMS/IVR, teleconsultation, maps). Offline PWA storage sits between the user and synchronization layer.

## 15. DATABASE — MVP

**Tables:** users, patients, facilities, services, availability, referrals, referral\_events, followups, triage\_rules, notifications, audit\_logs.

**Relationship:** Patient → Referral → Source/Destination Facility → Referral Events → Follow-up.

## 16. API SURFACE — MVP

**Auth:** POST /api/auth/login | **Patients:** POST /api/patients; GET /api/patients/:id | **Triage:** POST /api/triage

**Facilities:** GET /api/facilities; GET /api/facilities/:id/availability | **Referrals:** POST /api/referrals; GET /api/referrals; GET /api/referrals/:id

**Actions:** PATCH /api/referrals/:id/accept, /reject, /arrive, /complete | **Follow-up:** POST/GET /api/followups | **Dashboard:** GET /api/dashboard/stats

## 17. SECURITY & PRIVACY

Use secure authentication, role-based access control, HTTPS, password hashing, input validation, API authorization, audit logs, data minimization and appropriate encryption. Patient information should only be exposed to authorized users for the relevant workflow.

Health-data sharing should use appropriate consent and approved interoperability workflows. Do not claim production ABDM access unless actually obtained.

## 18. AI / INTELLIGENCE SCOPE

Use explainable **rules-based triage** in the MVP. Voice and OCR are assistance features. Healthcare staff confirm information and the final care pathway. Do not position Medi-Saarthi as an autonomous diagnosis system.

## 19. INTEROPERABILITY

Design for FHIR and approved ABDM workflows. During the hackathon, external government integrations can be represented through mock APIs/adapters if official production access is unavailable. Medi-Saarthi coordinates with existing systems rather than replacing them.

## 20. NON-FUNCTIONAL REQUIREMENTS

**Performance:** responsive dashboards/APIs. **Availability:** essential client workflows tolerate temporary network loss. **Security:** protected information and role-based access. **Scalability:** one district → multiple districts/state. **Maintainability:** modular services/adapters.

## 21. SUCCESS METRICS

Referral completion rate; missed referral rate; referral processing time; follow-up completion rate; availability verification rate; frontline-worker data-entry time compared with manual entry.

## 22. MVP VS FUTURE SCOPE

**MVP:** patient registration, basic voice/OCR demo, rule-based triage, facility readiness demo, Referral Guard, basic offline mode, follow-up and basic admin dashboard.

**Future:** advanced multilingual support, real facility integrations, advanced offline sync, advanced analytics, approved ABDM/eSanjeevani/108/102 integrations, production SMS/IVR and broader deployment.

## 23. HACKATHON DEVELOPMENT ORDER

**Sprint 1:** React/PWA, Node/Express, PostgreSQL, authentication, base UI.

**Sprint 2:** patient registration, facilities, referral creation/status.

**Sprint 3:** Pulse-Check, Referral Guard, rejection handling, follow-up.

**Sprint 4:** rules-based triage, voice, OCR, offline storage.

**Sprint 5:** dashboards, demo data, polishing, error handling and final demo.

## 24. TEAM DIVISION — 4 MEMBERS

**Member 1 — Frontend:** worker, referral, facility and admin interfaces.

**Member 2 — Backend:** APIs, authentication, referral engine and triage logic.

**Member 3 — Database/Infrastructure:** PostgreSQL, schema, Redis, offline synchronization and deployment.

**Member 4 — Intelligence/Integration:** voice, OCR, availability, notifications, mock FHIR/ABDM adapters and testing.

## 25. FINAL PRODUCT DEFINITION

**Medi-Saarthi is an offline-friendly rural healthcare coordination platform that connects frontline workers, healthcare facilities and patients through assisted intake, rule-based care navigation, facility readiness verification, closed-loop referral tracking and follow-up.**

**Core differentiator:** Existing systems may provide care, consultation or records. Medi-Saarthi coordinates what happens between them.

**MVP:** Register → Triage → Check readiness → Refer → Accept → Arrive → Complete → Follow-up.